Musanze Harvest Intelligence Service

Technical report - SWE 3513 Artificial Intelligence, Individual Assignment 2

Item	Value
Full name	Mohammed Bashier Ahmed
Registration number	25/27659
Institutional username	ug2527659@ines.ac.rw
Verification code	AI-A2-7659
Seed LLLL	7659
Institution	INES-Ruhengeri

1. Identity and decision problem

Stakeholder. The field officer of a potato cooperative in Musanze, who receives farm records in inconsistent spreadsheets and has unreliable internet access.

Prediction questions. (1) How many tons will a farm harvest (regression)? (2) Is the farm at high production risk, meaning its yield per hectare is below the median (classification)? (3) Which farms share similar conditions (clustering)?

Why local. Training and command-line prediction run offline from one command; a small FastAPI service allows later integration with a mobile application.

Ethical boundary. Classroom decision-support prototype trained on synthetic data. It does not replace an agronomist and must not be used to decide access to finance, insurance or public services. Human review is required.

Synthetic data. All records are generated from the formula in the assignment and the registration-based seed. No real farm or person is represented.

2. Data and reproducibility

Column	Generation rule	Role
rainfall_mm	Uniform 600-1400	Numeric feature
avg_temp_c	Uniform 12-26	Numeric feature
soil_moisture_pct	Uniform 20-80	Numeric feature
fertilizer_kg	Uniform 0-250	Numeric feature
field_size_ha	Uniform 0.2-4.0	Numeric feature
season	Random A or B	Categorical feature
cooperative_support	Random 0 or 1	Binary feature
yield_tons	Assignment formula, noise SD 0.80, minimum 0.50	Regression target
high_risk	Yield per hectare below its median	Classification target

Personal seed rule. The seed is the last four digits of the registration number, LLLL = 7659. The generator uses numpy.random.default_rng(7659); the base row count is 240 + (7659 mod 41) = 273. run_all.py rejects any other seed.

Row count and deliberate problems. With m = round(0.05 x 273) = 14, the generator removed 14 values of soil_moisture_pct and 14 values of fertilizer_kg and appended 3 exact duplicates. The CSV has 276 rows; the validator found 14 and 14 missing values and 3 duplicate rows, and the pipeline fails if these counts differ from the generator rules (this detects manual edits). Duplicates are dropped, leaving 273 rows; missing values are imputed inside the pipeline after the split.

SHA-256 fingerprint of the final CSV bytes (stored in artifacts/data_fingerprint.txt):

`f6536a25a3cebf263fdd9117db03b9debc91f11b3cc55560672ea7c721429649`

Re-running run_all.py twice reproduced the same fingerprint and byte-identical metric files, prediction table and fingerprint file.

3. Pipeline design

Split. After removing duplicates, a stratified (on high_risk) split with test size 0.25 and random_state=7659 gives 204 training rows and 69 test rows. The test set is used only for final reporting.

Preprocessing (one ColumnTransformer, fitted on training rows only). Numeric features: median imputation then standard scaling. season: one-hot encoding with unknown categories ignored. cooperative_support: passed through. The same fitted transformer is embedded in each saved model Pipeline, so training and inference apply identical transformations, and a copy is saved as models/preprocessor.joblib.

Leakage controls. yield_tons, yield per hectare and high_risk are never inputs to classification (assert_no_leakage runs before training and a test checks it). Imputers, scalers and encoders are never fitted before the split (a test confirms the imputer medians equal the training medians). Clustering uses no target.

Model selection. Candidates are compared with 5-fold cross-validation on the training set only (RMSE for regression, F1 for classification); the baseline is reported but cannot be selected. Test-set scores are reported for every model but do not choose the winner.

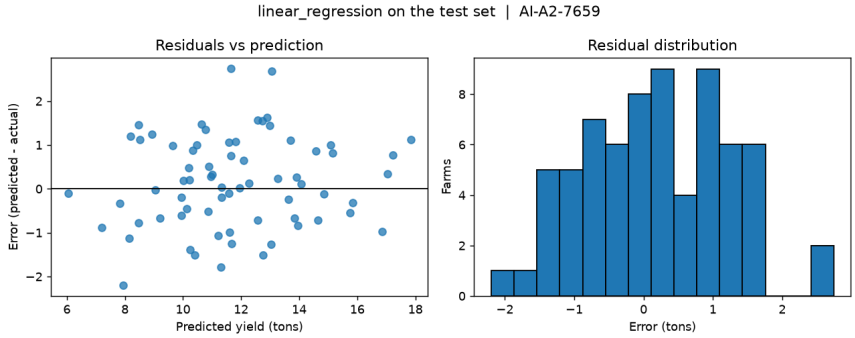
Task	Baseline	Candidates	Selection metric
Regression	Mean predictor	Linear regression, random forest, gradient boosting	CV RMSE (lowest)
Classification	Majority class	Logistic regression, random forest, gradient boosting	CV F1 (highest)
Clustering	-	K-means, K = 2 to 6 on scaled features	Silhouette (highest)

All random states are set to the seed; models use a single thread, which keeps results deterministic.

4. Regression and classification

Regression - held-out test set (selected: linear_regression)

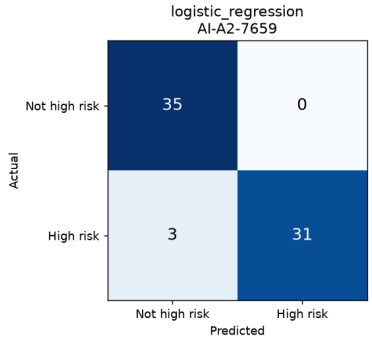
Model	CV RMSE (train)	Test MAE	Test RMSE	Test R2
mean_baseline	2.7487	2.1814	2.6347	-0.0216
linear_regression *	1.0283	0.8515	1.0408	0.8406
random_forest	1.4340	1.1809	1.4148	0.7054
gradient_boosting	1.3060	1.0709	1.2967	0.7526



Important regression error (own interpretation). For test row 4 the model predicted 11.65 t, but the real yield was 8.91 t, so the prediction was too high by 2.74 t. This is the largest error in the test set. The data contains random noise that no model can predict, so some farms will always be far from the prediction.

Classification - held-out test set (selected: logistic_regression)

Model	CV F1	Precision	Recall	F1	ROC-AUC	Confusion [[TN FP][FN TP]]
majority_baseline	0.262	0.000	0.000	0.000	0.500	[[35, 0], [34, 0]]
logistic_regression *	0.960	1.000	0.912	0.954	0.993	[[35, 0], [3, 31]]
random_forest	0.911	0.964	0.794	0.871	0.937	[[34, 1], [7, 27]]
gradient_boosting	0.897	1.000	0.765	0.867	0.986	[[35, 0], [8, 26]]

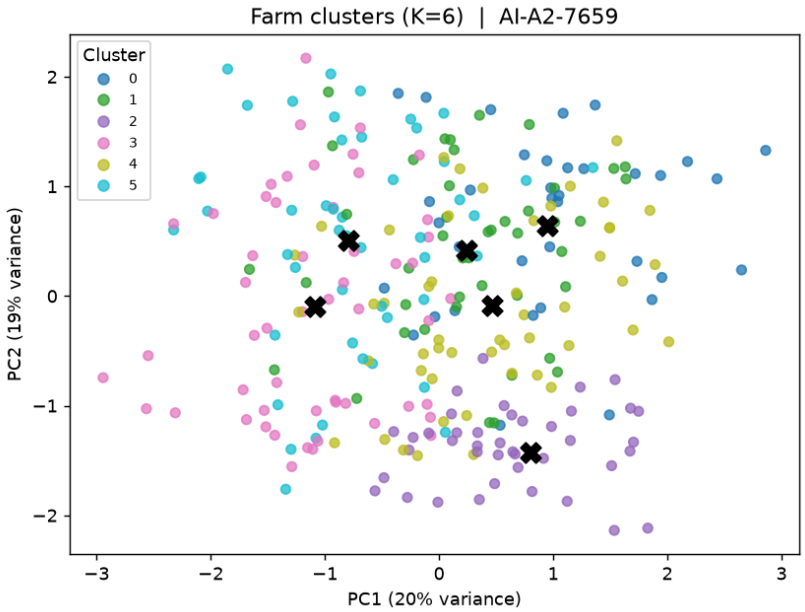


Important classification error (own interpretation). The model missed 3 high-risk farms (rows 99, 4 and 119). Their predicted probabilities (0.30, 0.12 and 0.35) were below 0.5, so they were labelled as not high risk. These farms are probably close to the median yield-per-hectare limit, where small random changes can flip the label. There were no false alarms.

5. Clustering

Features were imputed and scaled with the same preprocessing; K-means was fitted on training rows for K = 2 to 6. K=6 has the highest silhouette score (0.1504). The silhouette scores are low, so the data has weak natural cluster structure; this is expected because the synthetic features were drawn independently. Clusters are useful as practical farm groupings, not as natural categories.

K	2	3	4	5	6
Silhouette	0.1429	0.1305	0.1368	0.1459	0.1504



Cluster	Farms (train)	Profile
0	31	Farms with heavy fertilizer, large fields, dry soil, mostly season B.
1	31	Farms with small fields, moist soil, high rainfall.
2	30	Farms with low rainfall, warm temperature, moist soil.
3	41	Farms with cool temperature, small fields, low rainfall.
4	39	Farms with large fields, moist soil, little fertilizer.
5	32	Farms with little fertilizer, dry soil, high rainfall.

Farm cluster (own interpretation). Cluster 3 is the largest group (41 farms): cool temperature, small fields and low rainfall. Officials can use these groups to see quickly which farms have similar problems and plan solutions for them together. The groups are not sharply separated (silhouette about 0.15), so they are only a rough guide.

6. API and testing

Endpoint	Purpose	Success	Failure
GET /health	Service status and verification code	200	-
GET /model-info	Seed, fingerprint, selected models, test metrics	200	503 if models are missing
POST /predict	Typed farm record to yield, risk, probability, cluster	200	422 for missing field, wrong type, out-of-range or unknown field

Valid request (samples/valid.json):

```
{"rainfall_mm": 1000.0, "avg_temp_c": 18.5, "soil_moisture_pct": 55.0, "fertilizer_kg": 120.0,
"field_size_ha": 1.5, "season": "A", "cooperative_support": 1}
```

Invalid request (samples/invalid.json) has rainfall 2500, temperature "warm", season "C" and no field_size_ha; the CLI lists all four problems and exits with code 2, and the API returns HTTP 422 with the details. The request model uses strict types, so a numeric string is rejected rather than silently converted.

Tests. tests/test_system.py contains 10 pytest tests: valid inference, missing field, wrong type, range error, reproducibility and fingerprint, dataset signature with deliberate problems, wrong seed, leakage guard, train-only preprocessing, and an API test covering /health, /model-info and valid and invalid /predict calls. Failure paths are tested, not only the happy path. The result of the final run is recorded in evidence/AI_A2_2527659_TEST_LOG.pdf.

Failure behaviour. If models are missing, the CLI returns a message telling the user to run run_all.py and the API returns 503; it never returns a guessed prediction.

7. Model card and reflection

Intended use. Classroom decision support for a field officer: rough yield estimates, a risk screen and farm grouping on synthetic data.

Prohibited use. Deciding loans, insurance, subsidies or any access to public services; replacing an agronomist; use on real farms without retraining and validation on real data.

Limitations. The data is synthetic and follows a known formula, so high scores say little about real farms. The sample is small (69 test farms), so metrics carry wide uncertainty. Inputs outside the training ranges are rejected, not extrapolated. Clusters are weak groupings.

Risks. Over-trust in a confident probability; a false negative that hides a struggling farm; fairness problems if cooperative_support were later used to ration resources.

Human oversight. Every prediction is advice for a person to review; the API response repeats the disclaimer.

Reflection. The simplest model, linear regression, won the regression task because the generator is almost linear, and logistic regression won classification; more complex models did not help on 204 training rows. With real data I would collect more seasons, check measurement quality, calibrate probabilities, evaluate errors by district and validate with agronomists before any use.